

Debrief: Motion Does Not Need Maintenance

Resistance explains why everyday objects slow down

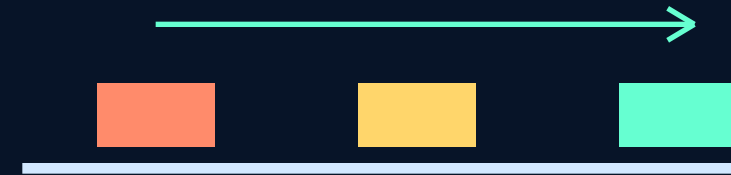
Evidence

- Less friction produced longer motion.
- Air track motion was nearly uniform.

Core Claim

- Motion changes when net force is nonzero.
- Balanced forces do not change velocity.

Visual Model



Less resistance -> longer distance

Newton's First Law

The rule for rest and uniform straight line motion

Law Statement

- Objects remain at rest or move uniformly
- unless a nonzero net force acts on them.

Important Detail

- Forces can exist and still be balanced.
- Net force, not any single force, changes motion.

Visual Model

NET FORCE = 0

Rest stays rest

Uniform straight-line motion stays uniform

Inertia

The tendency to keep the current motion state

Definition

- Inertia is the property of maintaining motion state.
- Every object has inertia.

Mass Connection

- Greater mass means greater inertia.
- More difficult to start, stop, or turn.

Visual Model



A push starts motion.

Experiment Conclusions

Translate observations into physics language

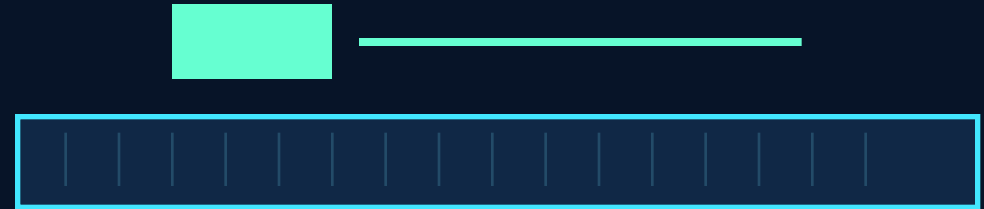
Surface Test

- Rough surface: larger resistance.
- Smooth surface: smaller resistance.

Air Track

- Nearly zero resistance.
- Nearly uniform straight-line motion.

Visual Model



Nearly zero net force $\rightarrow$ nearly uniform motion

Exit Ticket

Answer before leaving the mission

Question 1

- A puck glides at constant speed. What is the net force?
- Explain using Newton's First Law.

Question 2

- Why do real objects usually slow down?
- Name the force and its effect.

Visual Model

NET FORCE = 0

Rest stays rest

Uniform straight-line motion stays uniform